Elvis F. Elli

From: Agricultural & Environmental Letters <onbehalfof@manuscriptcentral.com>
Sent: Friday, May 8, 2026 12:26 PM
To: Elvis F. Elli
Cc: Elvis F. Elli; clsantos@iastate.edu; Samuel B Fernandes; msalmeron@uky.edu; battisti@ufg.br; rogerio.noia@inrae.fr; rob.malone@ars.usda.gov; Alan.Franzluebbers@usda.gov; ajfranzl@ncsu.edu; O'Brien, Peter - REE-ARS
Subject: Agricultural & Environmental Letters - Decision on Manuscript AEL-2026-04-0047-RL

08-May-2026

Dear Dr. Elvis Elli,

Thank you for submitting your manuscript, Earlier planting and late-maturing soybeans offset high temperature stress (AEL-2026-04-0047-RL), for publication in Agricultural & Environmental Letters. In view of the criticisms of the reviewer(s) found at the bottom of this letter, your manuscript has been denied publication. Although you are no doubt disappointed with the decision to decline publication, I appreciate the opportunity to consider this paper and encourage you to continue to submit your research results to Agricultural & Environmental Letters.

Thank you for choosing Agricultural & Environmental Letters as an outlet for your work.

Sincerely,

Dr. Peter O'Brien
Technical Editor, Agricultural & Environmental Letters

Editor: Dr. Alan Franzluebbers, Alan.Franzluebbers@usda.gov, ajfranzl@ncsu.edu Associate Editor: Dr. Robert Malone, rob.malone@ars.usda.gov

Reviewer(s)' Comments to Author:
Reviewer: 1

Comments to the Author
The authors have conducted a simulation study to evaluate impacts of earlier planting and variety selection for soybean in the Mid-South to offset heat and drought stress. The paper is short with minimal content. Similar results and conclusions have been found in other literature, so the novelty and impact of the paper is minimal. Authors could perhaps work to expand the message to increase the novelty of the paper. There is plenty of room for expansion to a fuller, more comprehensive research paper.

L54-64 – Minimize use of first person pronouns here and throughout.
L57 – There appears a conflicting logic with L47-49. It is stated that earlier planting shifts the crop cycle to cooler periods, reducing exposure to heat and drought. But here, the hypothesis proposes later maturity group soybeans, which would extend the growing season. Please clarify the reasoning for the hypothesis and for use of later maturity group soybean to avoid heat and drought.
L84 – What does it mean that the Cropland Datalayer was used to define the fields? Define what about the fields?
L110 – What does “advancing sowing date by 12 days” mean? Earlier or later?
L186 – Discuss the limitations of the study in terms of uncertainty in the model structure or model input data.

L194 – Overall, the paper is concise and content is minimal. The study provides new information for soybean in the Mid-South, but the conclusions are not particularly novel or earthshattering. How could the paper be expanded to be more impactful?

Reviewer: 2

Comments to the Author
The paper is well written and the objectives are clearly stated. The methods are well described and model calibration is based on sound data. Validation use a wide range of environments. The results are well presented. However, it would be interesting to learn a little more on the background of the yield effects, e.g., how temperature increase, early sowing and late maturity class affect the length of the seed filling period. Since soybean is responsive to short days, it would be good to know if APSIM considers any photoperiod effect and if this differs in the parameters of the different maturity groups.
Minor corrections:
line 156: please add "+" in front of "2°C scenario"
Supplement figure S3: please mention if these values are without adaptation measures.
The references of NASS (2023) and of Salmerón et al. 2016, which ae listed in the supplement reference list are never mentioned in the supplemwnt.

Reviewer: 3

Comments to the Author
The topic of the manuscript is interesting; however, the results lack novelty. The study emphasizes the need for actionable solutions to increasing temperatures in the U.S. Mid-South. To address this, a calibrated APSIM model is used to simulate soybean yield under future climate scenarios (+2°C increase in mean temperature and/or elevated CO_2), considering current and earlier planting dates in combination with changes in maturity group (MG).
While the topic is relevant, the findings are not new. Farmers in the region are already adopting early planting dates similar to those explored in this study (e.g., early May). Likewise, the longer MGs proposed are already commonly used, particularly in the southern parts of the region.

Additionally, the manuscript does not provide a clear explanation or mechanism underlying the simulated yield increases across scenarios. Important variables—such as the photothermal quotient during critical periods—are not explored. Evaluating more extreme scenarios and, importantly, providing a mechanistic explanation for the observed yield responses would strengthen the manuscript and improve its relevance for future readers.

Main comments
- Lines 50–53: This is an important point: early planting is not always optimal, as there is a planting window that maximizes radiation capture and varies with MG. However, only one “early” planting date is evaluated in this study. This does not allow identification or exploration of an optimal window, or at least this is not demonstrated.
- Line 64: Please include a brief introduction on the role and projected changes of CO_2, as this is currently missing.
- Lines 75–76: Is MG5 considered “late” in this context? Farmers are already planting MG5 and even MG6 in the southern parts of the region.
- Lines 110–113: Consider revising the second scenario description to:

“advancing the sowing date by 12 days, corresponding to an early but realistic planting window based on NASS planting progress data for the studied region (NASS, 2023)”.
The current reference to the “upper tail” suggests delayed planting. Also, please clarify what is meant by “realistic” (e.g., 10% vs. 30% planting progress). Would exploring even earlier, less common planting dates provide additional insight?
- Lines 156–158: When stating that “our study extends previous work,” please provide appropriate references.
- Lines 163–164: Clearly state what is novel in this study compared to similar simulations (e.g., Elli et al., 2022).
- Line 173: In discussing model uncertainties, please also consider uncertainties in phenology prediction. APSIM is somewhat limited mechanistically compared to other models—for example, it assumes a constant temperature response

throughout development, whereas in soybean this response changes across stages. This limitation and its potential impact on results should be discussed.

Supplementary material

- The optimization of phenology parameters is based on days, which may introduce bias. Using development rates would be more appropriate.
- Figure S1: Consider plotting days after sowing instead of day of year (DOY) to better evaluate model accuracy across environments, especially for earlier planting dates. It is unclear whether model performance under early planting was validated or simply assumed.

Additionally, differentiating key stages (R1, R5, R7) with distinct colors would improve clarity.

- The reported RMSE (nearly two weeks) after optimization seems high. How does this compare with similar studies?

Minor comments

- Lines 41–42: Please provide a supporting reference.
- Figure 1 caption: Clarify that all maturity groups (including late-maturing) also include the +2°C scenario.
- There appear to be two separate reference lists. Please check and correct.

Associate Editor's Comments to Author:
Associate Editor
Comments to the Author:
The reviewers found the 1) topic of the manuscript interesting and relevant, 2) paper well written, and 3) methods well described and model calibration based on sound data. However, the reviewers note that results and conclusions have been reported and early planting dates similar to those examined (e.g., early May) and the longer maturity groups proposed are already commonly used. So the novelty and impact of the paper is unclear and limited. Items to help strengthen study include: evaluating more scenarios, providing a mechanistic explanation for the observed yield responses, clarify novelty of study compared to similar studies/simulations (e.g., Elli et al., 2022).

Technical Editor's Comments to Author:
Technical Editor
Comments to the Author:
As the AE noted, the reviewers did not find fault with the rigor or soundness of this work. Rather, the concerns were primarily related to novelty and impact of the manuscript. With this in in mind, I suggest the authors consider reframing the context of the manuscript to better highlight the originality of the work and the knowledge gap that it is addressing. The topic is certainly of interest to the readership of AEL, so if the re-worked manuscript still falls within journal word limits, I would invite you to resubmit for another round of peer review.